Roop Sai Charan Kothamasu

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Professional Summary

Software engineer and M.S. Computer Science candidate building Python data pipelines, backend services and REST APIs, SQL databases, distributed systems, and agentic AI. Experienced in automated testing and AWS deployment.

Technical Skills

Programming: Python, Java, C++, SQL, Go, JavaScript, TypeScript, C
Backend & Databases: REST APIs, Node.js, Express.js, PostgreSQL, MySQL, MongoDB, SQLite, SQL Schema Design
Data & Distributed Systems: ETL, Data Pipelines, Data Validation, Data Provenance, Distributed Systems, Concurrency, Algorithms, Data Structures
Cloud & Engineering: AWS (EC2, S3, CloudWatch), Docker, CI/CD, Git, Linux, Unit & Integration Testing
AI & Security: LLMs, RAG, AI Agents, PyTorch, Mamba, JWT, Role-Based Access Control (RBAC)

Education

University of Florida — M.S. Computer Science, GPA: 4.0/4.0 — Gainesville, FL | Aug 2025 – Aug 2027
SRM University AP — B.Tech Computer Science — Andhra Pradesh, India | Mar 2021 – Mar 2025

Professional Experience

Software Engineer, TRACER Data Platform | University of Florida / McTrans IPPD — 2026 – Present

- Built a Python ETL pipeline to normalize 1.2M+ state DOT records into 2,493 freeway segments across 71 routes using configurable NC/FL JSON schema adapters.
- Implemented overlap resolution, AADT propagation, and data provenance across 371 corridors; validated 99.97% lane and 100% speed concordance.
- Implemented HCM 7th Edition segmentation (96.08% spatial accuracy), exported 72-column HCS staging CSVs, and collaborated with McTrans engineering stakeholders.

Research Assistant, Agentic AI for Chip Design | University of Florida ECE — 2026 – Present

- Developing an LLM-powered AI orchestrator to connect chip-design questions with electronic design automation (EDA) tools and engineering workflows.
- Building a retrieval-augmented generation (RAG) pipeline to collect, clean, and chunk semiconductor textbooks, papers, and design documentation.
- Created Python validation scripts to repair math expressions and preserve technical terminology in source chunks from 200+ books and 50+ lectures.

Independent Researcher, Autonomous Vehicle World Models | University of Florida — Mar – Sep 2026

- Trained PyTorch Mamba world models on 8,000+ driving episodes (6.4M+ samples) for five-second trajectory prediction at under 12 ms inference latency.
- Ran 15+ Mamba, Transformer, and LSTM ablations; reduced MSE 23% versus vanilla Mamba and physically implausible outputs 47% using kinematic constraints.

Student Assistant, Chatbot & Web Development | UF Reitz Union Game Room — Apr – Oct 2026

- Developed a staff/customer chatbot for 50+ FAQ, operating-procedure, and equipment troubleshooting intents; integrated it with the Game Room website.
- Delivered self-service support that reduced repetitive staff inquiries by 80% and accelerated question resolution 3×.

Special Project Intern, Backend & Blockchain | SRM University AP — Dec 2024 – Mar 2025

- Built 12 Node.js REST APIs and 3 Solidity contracts for patient-record sharing with MongoDB, JWT, and RBAC; tested 500+ Ganache transactions.

AWS Cloud Computing Intern | APSSDC — Jun – Aug 2023

- Deployed 3 AWS EC2 web apps with CloudWatch monitoring; automated CI/CD using CodePipeline and GitHub Actions, cutting deployment from 25 to under 4 minutes.

Technical Projects

MeetMint — AI Productivity Platform | Go, React, Gemini, SQLite, RAG — Mar – May 2026

- Built a Go/React meeting assistant with 8 REST endpoints for summaries, action items, and RAG Q&A over 500+ SQLite records.
- Wrote 35+ unit/integration tests (92% coverage); measured sub-2s retrieval and under 300 ms p95 API latency.

Distributed Systems — Chord, Gossip & Push-Sum | Gleam, Erlang/OTP — 2026

- Implemented Chord peer-to-peer lookup with consistent hashing and finger tables across 10,000+ nodes (6.5 average hops).
- Implemented Gossip and Push-Sum protocols and evaluated convergence with 10,000+ concurrent Erlang/OTP actors.

Music Streaming Platform | Next.js, React, Stripe, PostgreSQL, Supabase — Jul – Sep 2023

- Built a Next.js subscription app with Stripe billing, Supabase Auth, and a normalized 8-table PostgreSQL schema.
- Indexed SQL queries to reduce playlist fetch time by 60%; implemented Google/GitHub OAuth login.

Publications

- R.S. Charan et al., *An Explainable AI Driven Hybrid Model for Enhanced Intrusion Detection in Network Security* (2025).
- R.S. Charan et al., *Analyzing Lossless Image Compression Techniques for IoT Devices* (ICEEICT 2024).